SRICHARAN AITHARAJU

srichar anaitharaju@gmail.com | (+91) 9059071512

/sricharan-aitharaju https://github.com/SriCharanAitharaju Hyderabad,India

PROFESSIONAL SUMMARY

Aspiring Electronics & Communication Engineering 3rd Year student with a strong goal to build a career in VLSI design. Proficient in Verilog, Cadence, Multisim, MATLAB, and embedded systems (Arduino, ESP32). Strong programming foundation in C, Java, and Python with hands-on academic projects in IoT and signal processing. Passionate about circuit design, digital systems, and semiconudctor technologies. Active leader and team player with experience as Joint Secretary of Kriya (2025-Present).

EDUCATION

Electronics And Communication Engineering|Anurag University | Hyderabad — **CGPA: 8.25 (Sem 3) | 2024 - Present**

XII | Nano Junior College | Nallakunta |Hyderabad — **80.2% | 2023**

X | Kamala Memorial High School |Nacharam| Hyderabad — **95.00% | 2021**

SKILLS

VLSI & Digital Design: Verilog HDL, RTL Design (Basic), Digital Logic Design, Combinational & Sequential Circuits, FSM Design (Basic), FPGA Fundamentals (Vivado)

EDA Tools: Cadence (Basics), Xilinx Vivado, Multisim

Programming : C, Java, Python

Embedded Systems & IoT: Arduino**,** ESP32**,** Sensor Interfacing

Core Electronics Concepts: Semiconductor Fundamentals, CMOS Basics, Circuit Design, Timing Concepts (Setup/Hold – Basic Understanding**)**

Relevant Exposure: Basic understanding of Digital Design Flow, Simulation & Verification Concepts

PROJECTS

- **JalNetra – Smart Water Monitoring System (SIH 2025)**

  Developed an IoT-based system to monitor water quality, flow, and usage in real time.Integrated sensors with Arduino for continuous data acquisition and processing Implemented basic logic for threshold detection and alert mechanisms Selected for Smart India Hackathon (SIH) 2025, demonstrating project relevance and scalability

- **Smart Railway Track Crack Detection System**

  Designed a prototype system using IR sensors and Arduino for real-time fault detection.
  Built a detection mechanism to identify cracks and trigger alerts.
  Focused on reliability and safety critical monitoring,simulating real world railway conditions.
  Demonstratedunderstanding of sensor interfacing and embedded system design

- **Fingerprint Quality Estimator (MATLAB)**
  - Developed an image processing model to evaluate fingerprint clarity and quality
  - Applied basic signal/image processing techniques using MATLAB
  - Improved biometric reliability by identifying low-quality inputs
  - Gained exposure to data analysis and algorithm-based evaluation

- **Hardware-Accelerated Edge-AI Plant Stress Monitoring System Zynq-(7000 FPGA | Verilog RTL)**
  - Designed an edge-based plant monitoring system using sensor data acquisition
  - Implemented Verilog RTL modules for data processing on FPGA
  - Utilized Zynq-7000 (PS + PL) for hardware-software integration
  - Achieved real-time processing using hardware acceleration
  - Worked with Vivado design flow (simulation & synthesis)

EXPERIENCE & LEADERSHIP

- **Joint Secretary – Kriya (Event Management Club) Anurag University | March 2026 – Present**

  Led execution of university events by coordinating teams and managing operations efficiently

- **Member – VLSI Club Anurag University | 2025 – Present**

  Engaged in workshops and projects on digital design, RTL concepts, and FPGA workflows.

- **Member – NSS (National Service Scheme) Anurag University | 2025 – Present**

  Participated in community service initiatives and collaborative social impact activities.

ACHIEVEMENTS

- **Qualified JEE Advanced (2023)**
- **Selected for Smart India Hackathon 2025 for JalNetra – Smart Water Monitoring System**
- **Co-organized large-scale university technical events including Sakshi's Spell Bee & Math Bee (2026)**

CERTIFICATIONS

- **Arduino & Embedded Systems Workshop**
  Hands-on training in microcontrollers, sensor interfacing, and real-time applications
- **Cadence EDA Tools Workshop**
  Introduction to VLSI design flow and EDA tools usage